

Ryoichiro Agata

RESEARCHER

Japan Agency for Marine-Earth Science and Technology (JAMSTEC), Yokohama, Kanagawa, Japan

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Professional Experience

2020-	Researcher , Japan Agency for Marine-Earth Science and Technology (JAMSTEC), Japan
2017-2020	Postdoctoral Researcher , Japan Agency for Marine-Earth Science and Technology (JAMSTEC), Japan
2016-2017	JSPS Research Fellowship for Young Scientists PD , Japan Society for the Promotion of Science, Japan
2016-2017	Research fellow , Earthquake Research Institute, University of Tokyo, Japan

Education

The University of Tokyo DOCTOR OF ENGINEERING • Advisor: Prof. Tsuyoshi Ichimura	<i>Tokyo, Japan</i> 2013-2016
The University of Tokyo MASTER OF ENGINEERING • Advisor: Prof. Tsuyoshi Ichimura	<i>Tokyo, Japan</i> 2012-2013
The University of Tokyo BACHELOR OF ENGINEERING • Advisor: Prof. Tsuyoshi Ichimura	<i>Tokyo, Japan</i> 2008-2012

Publications

REFEREED JOURNAL, FIRST-AUTHORED ETC.

Ryoichiro Agata, Satoru Baba, Ayako Nakanishi, Yasuyuki Nakamura: Rapid Hypocenter Determination Tool for the Nankai Trough Subduction Zone Using Physics-Informed Neural Networks. *Seismological Research Letters*, 2025. doi: <https://doi.org/10.1785/0220240377>

Ryoichiro Agata: Quantification of Uncertainty and Its Propagation in Seismic Velocity Structure and Earthquake Source Inversion, *Journal of the Seismological Society of Japan*. 2nd ser, 2025. Published in Japanese and English preprint available: <https://arxiv.org/abs/2411.17997>

Ryoichiro Agata, Kazuya Shiraishi, Gou Fujie: Physics-informed deep learning quantifies propagated uncertainty in seismic structure and hypocenter determination, *Scientific Reports*, 15, 1846, 2025. <https://doi.org/10.1038/s41598-024-84995-9> (open access).

Ryoichiro Agata, Kazuya Shiraishi, Gou Fujie: Bayesian seismic tomography based on velocity-space Stein variational gradient descent for physics-informed neural network, *IEEE Transactions on Geoscience and Remote Sensing*, vol. 61, pp. 1-17, Art no. 4506917, 2023. doi: 10.1109/TGRS.2023.3295414

Ryoichiro Agata, Nakata, R., Kasahara, A., Yagi, Y., Seshimo, Y., Yoshioka, S., & Iinuma, T.: Bayesian Multi-Model Estimation of Fault Slip Distribution for Slow Slip Events in Southwest Japan: Effects of Prior Constraints and Uncertain Underground Structure. *Journal of Geophysical Research: Solid Earth*, 127(8), e2021JB023712, 2022.

Ryoichiro Agata, Amato Kasahara and Yuji Yagi: A Bayesian inference framework for fault slip distributions based on ensemble modeling of the uncertainty of underground structure - With a focus on uncertain fault dip, *Geophysical Journal International*, 225(2), 1392-1411, 2021.

Takane Hori*, **Ryoichiro Agata***, Tsuyoshi Ichimura, Kohei Fujita, Takuma Yamaguchi, Takeshi Iinuma: High Fidelity Elastic Green's Functions for Subduction Zone Models Consistent With the Global Standard Geodetic Reference System, *Earth, Planet and Space*, 73(1), 1-12, 2021. (* Equal contributors)

- Ryoichiro Agata**, Takane Hori, Sylvain D. Barbot, Mamoru Hyodo and Tsuyoshi Ichimura: Quasi-static Simulation Method of Earthquake Cycles Based on Viscoelastic Finite Element Modeling. In: Itou H., Hirano S., Kimura M., Kovtunen V., Khludnev A. (eds) *Mathematical Analysis of Continuum Mechanics and Industrial Applications III. CoMFoS 2018. Mathematics for Industry*, vol 34, 2020. Springer, Singapore. https://doi.org/10.1007/978-981-15-6062-0_11.
- Ryoichiro Agata**: Introduction of covariance components in slip inversion of land and seafloor geodetic data and application to slip deficit rate estimation in the Nankai Trough subduction zone, *Geophysical Journal International*, 221(3):1832–1844, 2020, doi:10.1093/gji/ggaa116.
- Ryoichiro Agata**, Sylvain D. Barbot, Kohei Fujita, Mamoru Hyodo, Takeshi Iinuma, Ryoko Nakata, Tsuyoshi Ichimura, and Takane Hori: Rapid mantle flow with power-law creep explains deformation after the 2011 Tohoku mega-quake, *Nature Communications*, 10(1), 1385, 2019.
- Ryoichiro Agata**, Takane Hori, Keisuke Ariyoshi and Tsuyoshi Ichimura: Detectability analysis of interplate fault slips in the Nankai subduction thrust using seafloor observation instruments, *Marine Geophysical Research*, 2019, <https://doi.org/10.1007/s11001-019-09380-y>.
- Ryoichiro Agata**, Tsuyoshi Ichimura, Takane Hori, Kazuro Hirahara, Chihiro Hashimoto and Muneo Hori: An adjoint-based simultaneous estimation method of the asthenosphere's viscosity and afterslip using a fast and scalable finite element adjoint solver, *Geophysical Journal International*, 213(1), 461–474, 2018.
- Ryoichiro Agata**, Tsuyoshi Ichimura, Kazuro Hirahara, Mamoru Hyodo, Takane Hori and Muneo Hori: Robust and portable capacity computing method for many finite element analyses of a high-fidelity crustal structure model aimed for co-seismic slip estimation, *Computers & Geosciences*, 94, 121–130, 2016.
- Ryoichiro Agata**, T. Ichimura, K. Hirahara, M. Hyodo, T. Hori, C. Hashimoto and M. Hori: Numerical Verification Criteria for Coseismic and Postseismic Crustal Deformation Analysis with Large-scale High-fidelity Model, *Procedia Computer Science*, 51, 1534–1544, 2015. (Presented at International Conference on Computational Science 2015, Reykjavik, Iceland)
- Ryoichiro Agata**, T. Ichimura, K. Hirahara, M. Hyodo, T. Hori and M. Hori: Several Hundred Finite Element Analyses of an Inversion of Earthquake Fault Slip Distribution using a High-fidelity Model of the Crustal Structure, *Procedia Computer Science*, 29, 877–887, 2014. (Presented at International Conference on Computational Science 2014, Cairns, Australia)
- Ryoichiro Agata**, T. Ichimura, M. Hyodo, T. Hori, K. Hirahara and M. Hori: Fundamental Research for Improving Fault Scenario –Development of a Method for Crustal Deformation Analysis Using High-Fidelity Three-Dimensional Crustal Structure–, *Journal of Japan Society of Civil Engineers, Ser. A1 (Structural Engineering & Earthquake Engineering (SE/EE))*, 69(4), I_767-I_776, 2013 (in Japanese with English abstract)

REFEREED JOURNAL, COAUTHORED

- Tomohisa Okazaki, Take Ito, Kazuro Hirahara, **Ryoichiro Agata**, Masayuki Kano, Naonori Ueda: Three-dimensional crustal deformation analysis using physics-informed deep learning. *Journal of Geophysical Research: Solid Earth*, 131(4), e2025JB032618.
- Shinji Yamashita, Yuji Yagi, Ryo Okuwaki, Kousuke Shimizu, **Ryoichiro Agata**, Yukitoshi Fukahata, Potency density tensor inversion of complex body waveforms with time-adaptive smoothing constraint, *Geophysical Journal International*, Volume 231, Issue 1, October 2022, Pages 91–107, <https://doi.org/10.1093/gji/ggac181>
- Tomita, F., Iinuma, T., **Ryoichiro Agata**, & Hori, T. (2021). Development of a Trans-Dimensional Fault Slip Inversion for Geodetic Data. *Journal of Geophysical Research: Solid Earth*, 126(5), e2020JB020991.
- Shinji Yamashita, Yuji Yagi, Ryo Okuwaki, Kousuke Shimizu, **Ryoichiro Agata**, Yukitoshi Fukahata: Consecutive ruptures on a complex conjugate fault system during the 2018 Gulf of Alaska earthquake. *Scientific Report*, 11, 5979, 2021.
- Masaru Nakano, Shane Murphy, **Ryoichiro Agata**, Yasuhiko Igarashi, Masato Okada, Takane Hori: Self-similar stochastic slip distributions on a non-planar fault for tsunami scenarios for megathrust earthquakes. *Progress in Earth and Planetary Science* 7, 45, 2020. <https://doi.org/10.1186/s40645-020-00360-0>
- Tsuyoshi Ichimura, **Ryoichiro Agata**, Takane Hori, Kenji Satake, Kazuto Ando, Toshitaka Baba and Muneo Hori: Tsunami Analysis Method with High-Fidelity Crustal Structure and Geometry Model: *Journal of Earthquake and Tsunami*, 1750018, 2017.
- Takuma Yamaguchi, Tsuyoshi Ichimura, Yuji Yagi, **Ryoichiro Agata**, Takane Hori and Muneo Hori: Fast crustal deformation computing method for multiple computations accelerated by a graphics processing unit cluster. *Geophysical Journal International*, 210(2), 787–800, 2017.

Takuma Yamaguchi, **Ryoichiro Agata**, Tsuyoshi Ichimura, Muneo Hori and Lalith Wijerathne: A Monte Carlo analysis method of crustal deformation computation on GPU clusters, Journal of Japan Society of Civil Engineers, Ser. A1 (Structural Engineering & Earthquake Engineering (SE/EE)), 73(4), 2017 (in Japanese with English abstract).

Tsuyoshi Ichimura, **Ryoichiro Agata**, Takane Hori, Kazuro Hirahara, Chihiro Hashimoto, Muneo Hori and Yukihiro Fukahata: An Elastic/Viscoelastic Finite Element Analysis Method for Crustal Deformation using a 3D Island-scale High-fidelity Model, Geophysical Journal International, 206(1), 114-129, 2016.

Tsuyoshi Ichimura, **Ryoichiro Agata**, Takane Hori, Kazuro Hirahara and Muneo Hori: Fast numerical simulation of crustal deformation using a three-dimensional high-fidelity model, Geophysical Journal International, 195 (3), 1730-1744, 2013.

Awards, Fellowships, & Grants

2024-2025	Grant-in-Aid for Transformative Research Areas A (Publicly Offered Research) , Japan Society for the Promotion of Science, Japan	4,030k JPY
2021-2023	Grant-in-Aid for Young Scientists , Japan Society for the Promotion of Science, Japan	3,770k JPY
2019-2020	Grant-in-Aid for Scientific Research on Innovative Areas (Publicly Offered Research) , Japan Society for the Promotion of Science, Japan	1,500k JPY
2014-2016	JSPS Research Fellowship for Young Scientists DC1 , Japan Society for the Promotion of Science, Japan	3,200k JPY

Other Research Experience

The Earth Observatory of Singapore, Nanyang Technological University	Singapore
HOST: PROF. SYLVAIN BARBOT	2016-2017
• Three-month visit for collaborative study (published as Agata+2019NatComms etc.)	

Outreach & Professional Development

PEER REVIEW

- Artificial Intelligence and Data Science
- Artificial Intelligence in Geosciences
- Communications Earth & Environment
- Engineering
- Geochemistry, Geophysics, Geosystems
- Geophysical Journal International
- Geophysical Research Letters
- Geosciences
- IEEE Transactions on Geoscience and Remote Sensing
- Journal of Geophysical Research: Machine Learning and Computation
- Journal of Geophysical Research: Solid Earth
- Surveys in Geophysics
- Mathematical Geosciences
- Progress in Earth and Planetary Science

PROFESSIONAL MEMBERSHIPS

- American Geophysical Union
- Japan Geoscience Union
- The Seismological Society of Japan
- The Japan Society for Industrial and Applied Mathematics